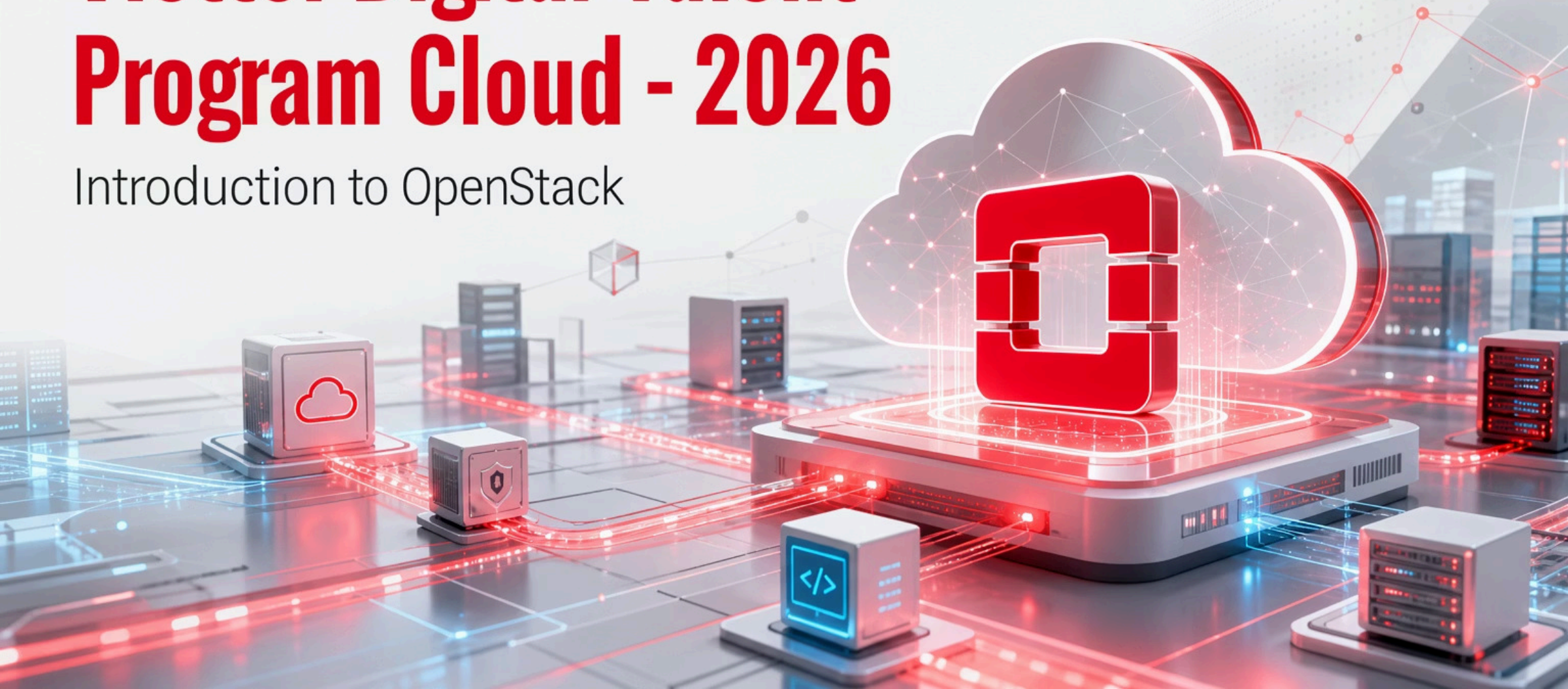


Viettel Digital Talent Program Cloud - 2026

Introduction to OpenStack



History and Development of OpenStack



2010

Initial ideas and
open-source community formed

2011 – 2012

OpenStack project
established

2013 – 2014

Key components developed
and integrated

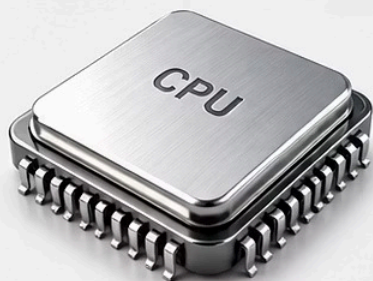
2015 – 2017

Rapid growth,
global adoption

2018 – Present

Mature platform,
expanding cloud and
container support

OpenStack by the Numbers



55 Million+

compute cores in
production worldwide.



5,000+

enterprises and mid-size
companies using
OpenStack.



27.55%

telecom market share,
for NFV and 5G.



**Walmart:
>1 Million cores**

across 10,000+ stores
worldwide.

OpenStack vs Kubernetes

Understanding the differences and complementary roles

OPENSTACK

Focus: Infrastructure Management (IaaS)

- Manages VMs, storage, and networking
- Provides bare-metal and virtual compute resources
- Components: Nova, Neutron, Cinder, Glance, Keystone
- Best for: Multi-tenant cloud infrastructure
- Hypervisor-based workloads

KUBERNETES

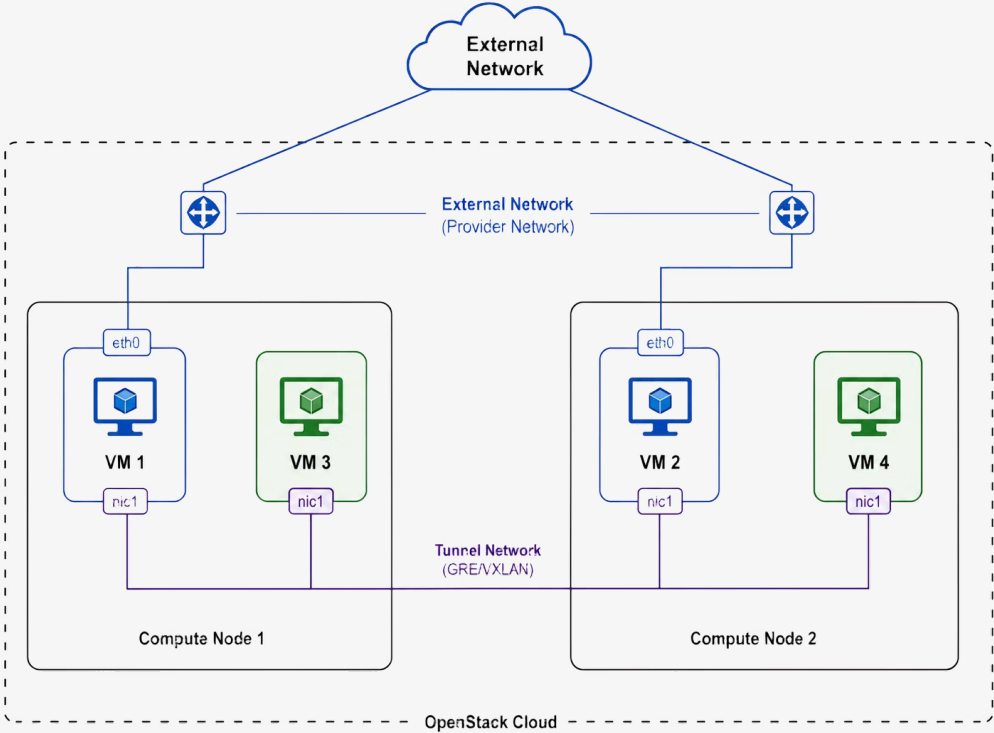
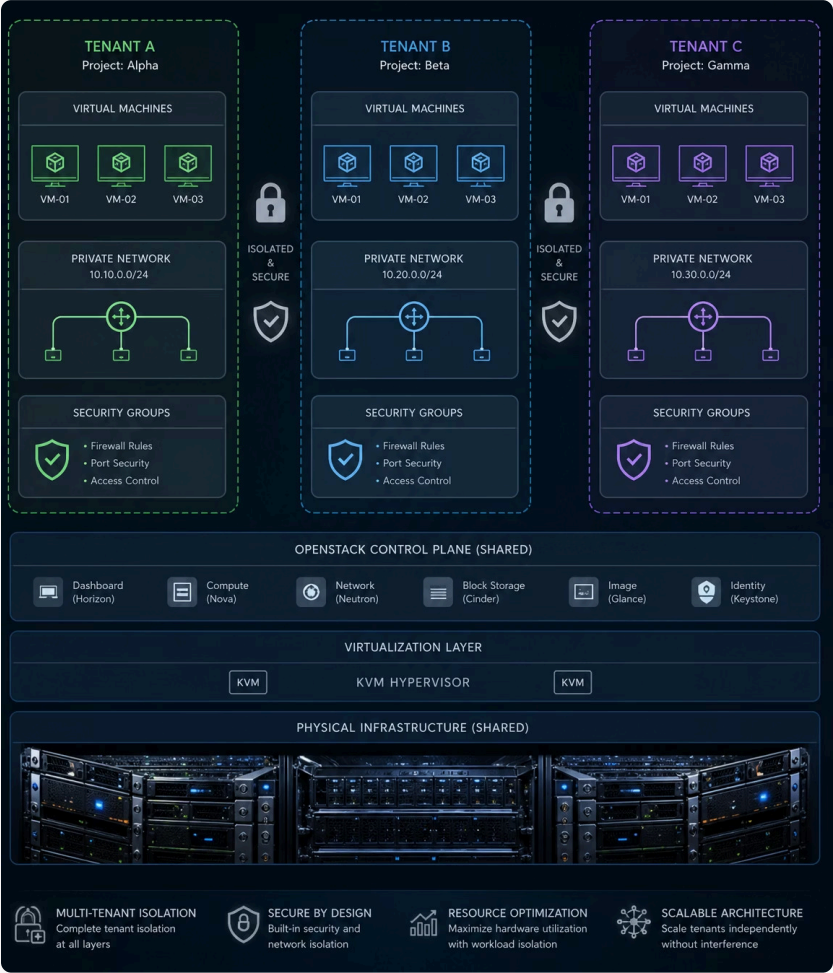
Focus: Container Orchestration (CaaS)

- Manages containers and microservices
- Auto-scaling, self-healing, rolling updates
- Components: Pods, Services, Deployments, Ingress
- Best for: Cloud-native applications
- Container-based workloads

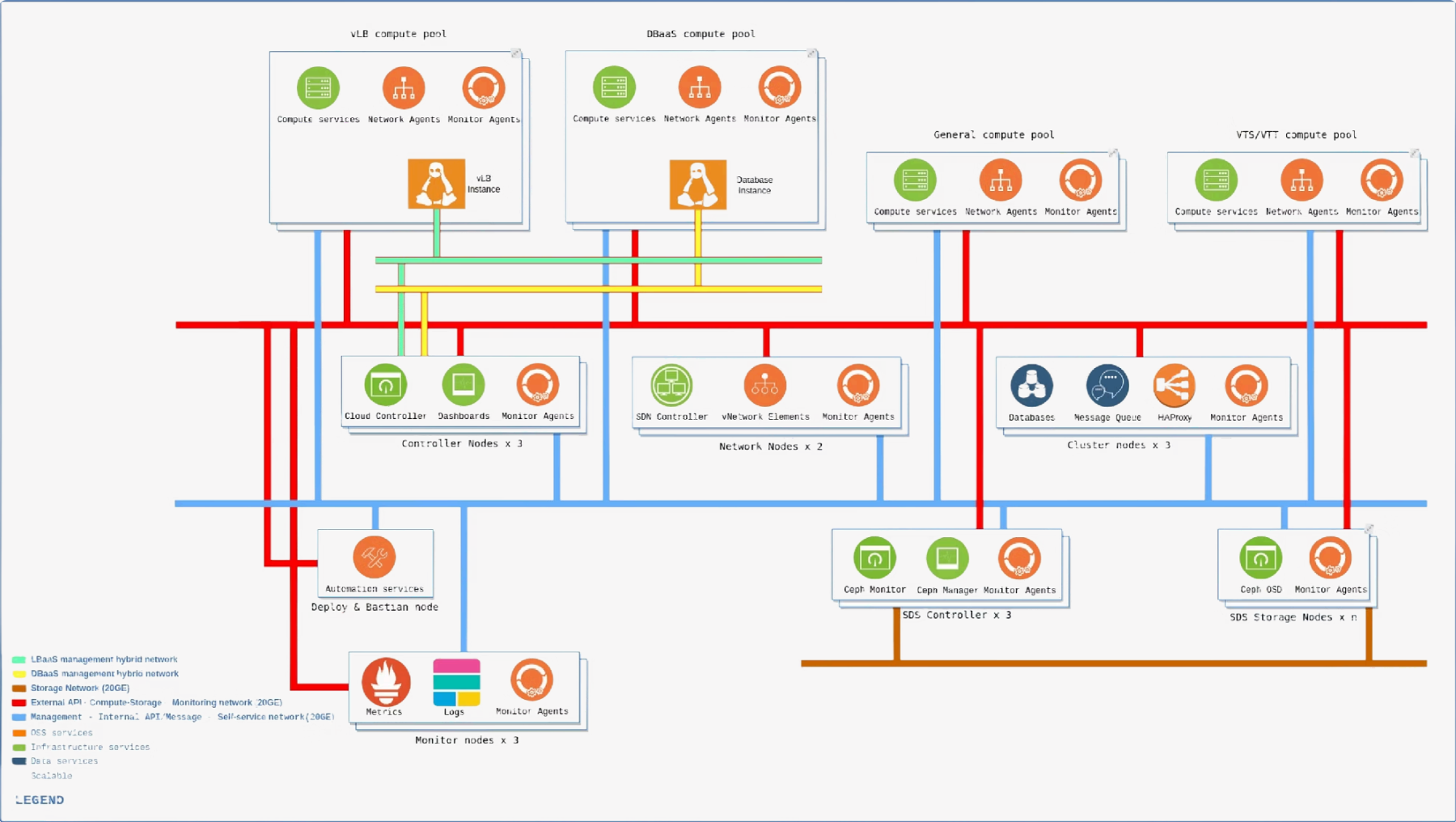
Better Together

OpenStack provides the infrastructure layer while Kubernetes orchestrates containerized workloads on top of it.

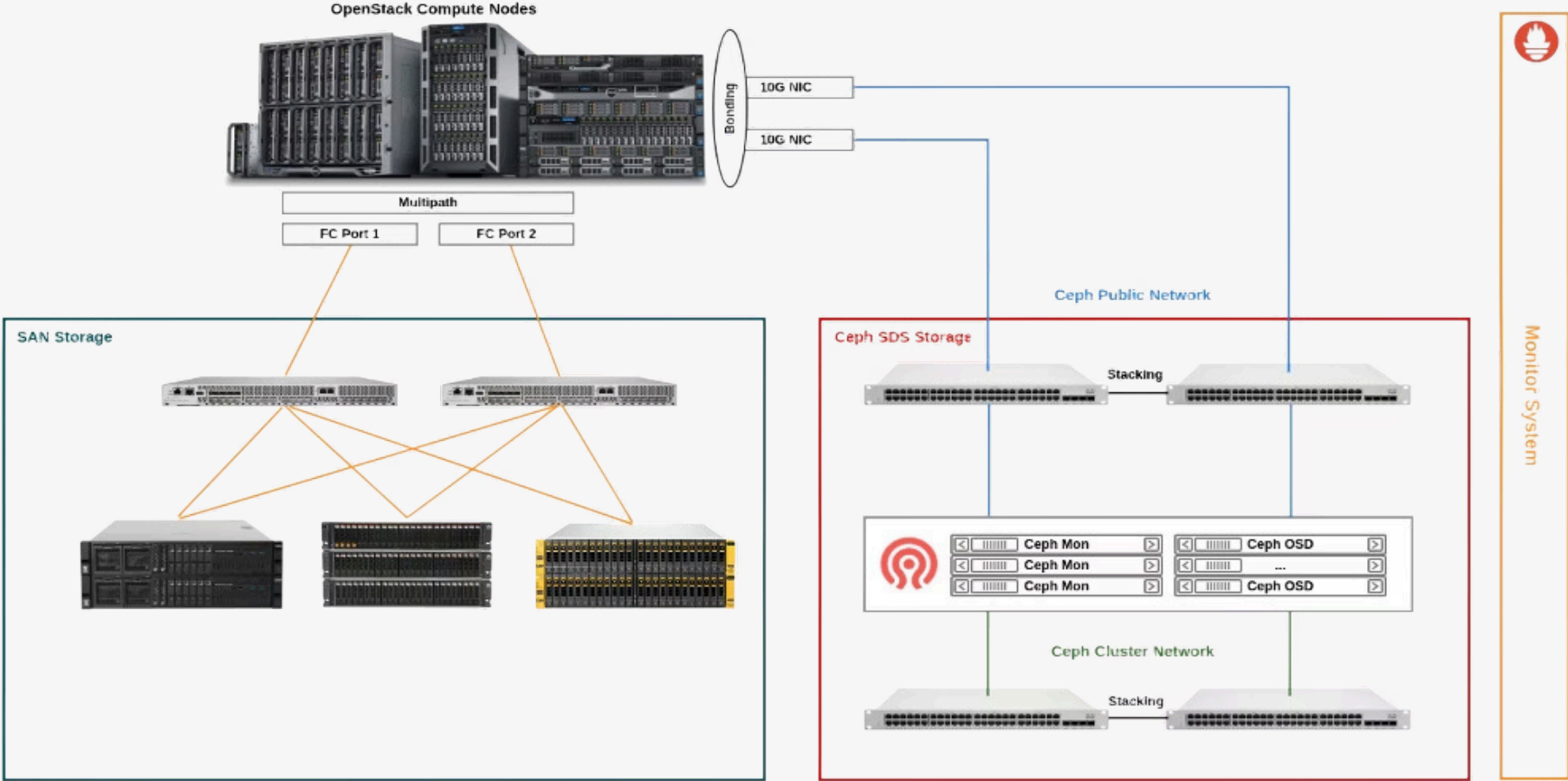
OpenStack Advantages



Deployment Architecture



Hardware In The Background



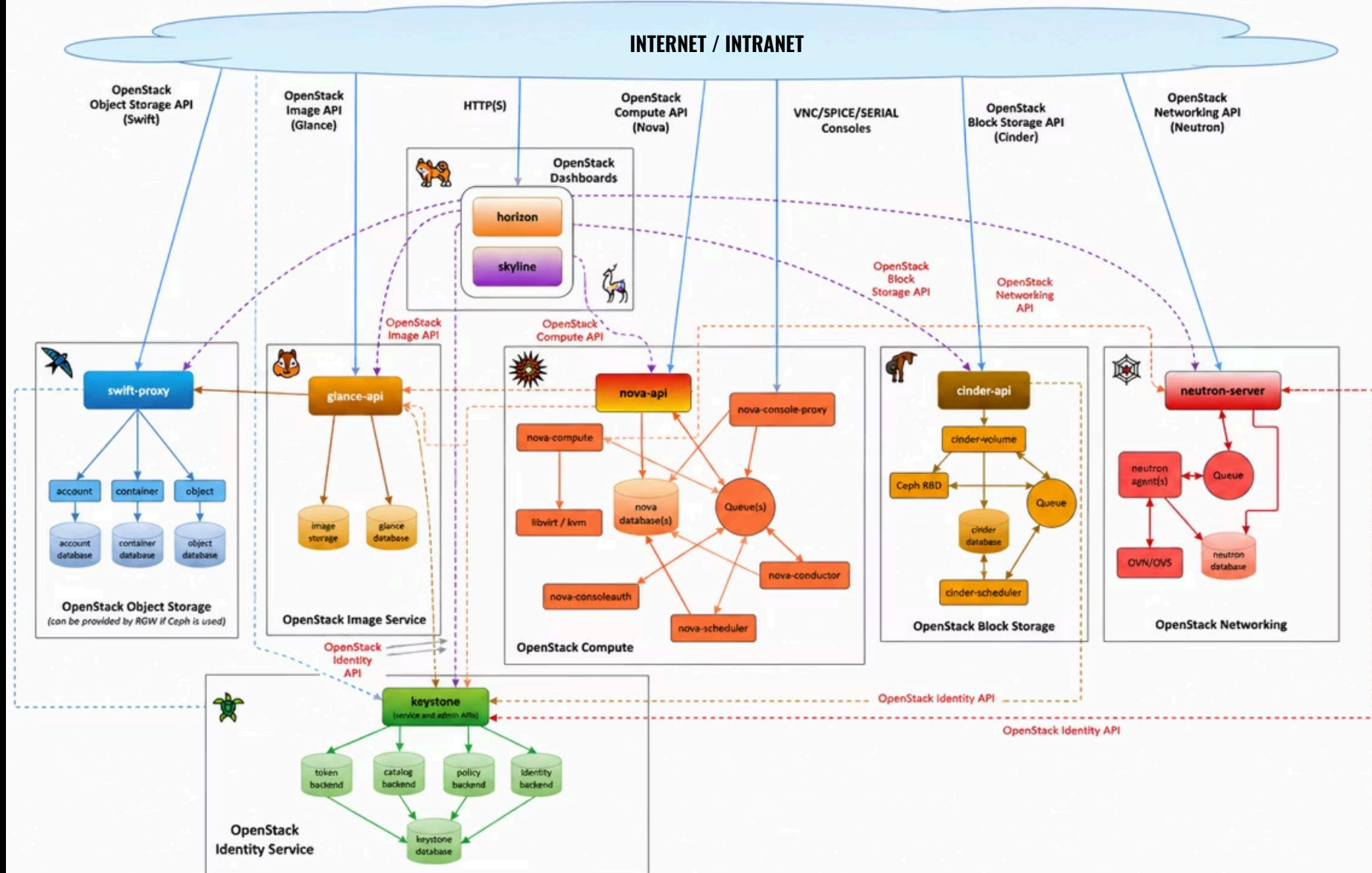
OpenStack Core Services



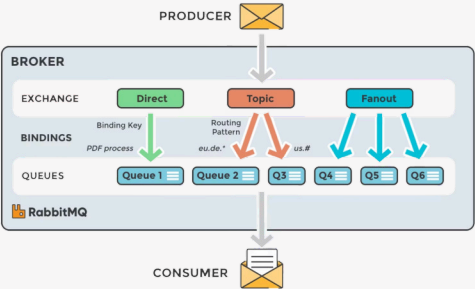
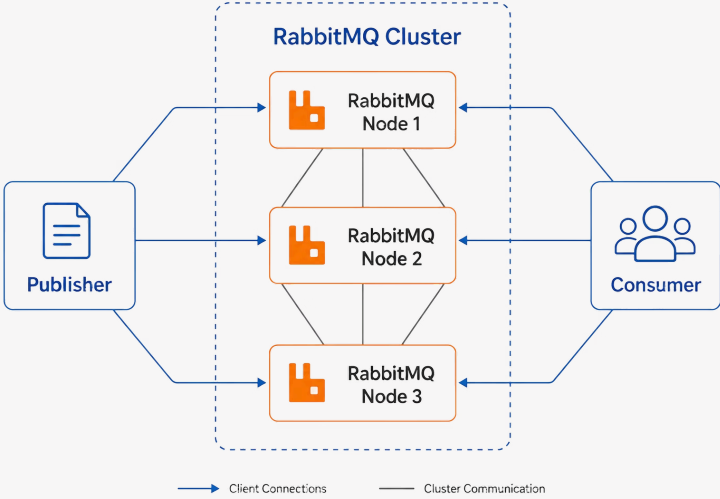
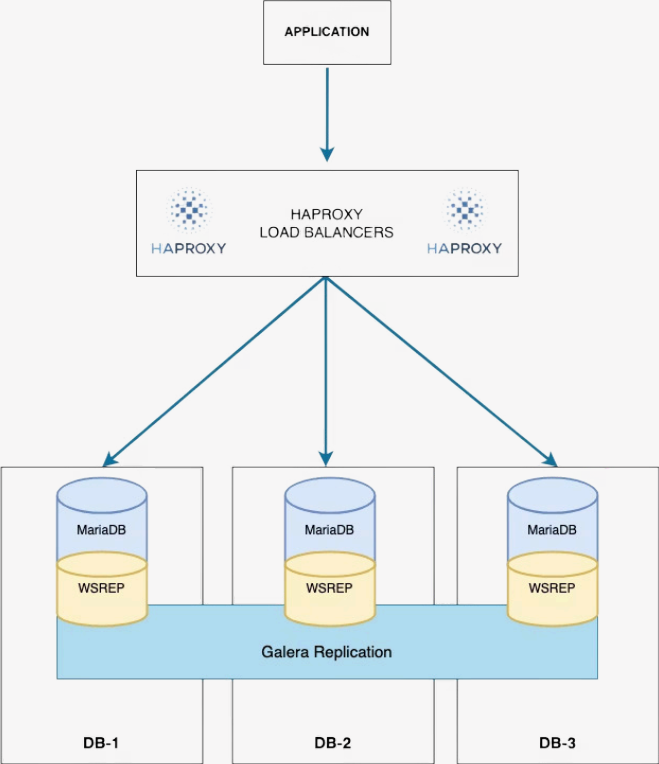
Web UI

OpenStack CLI

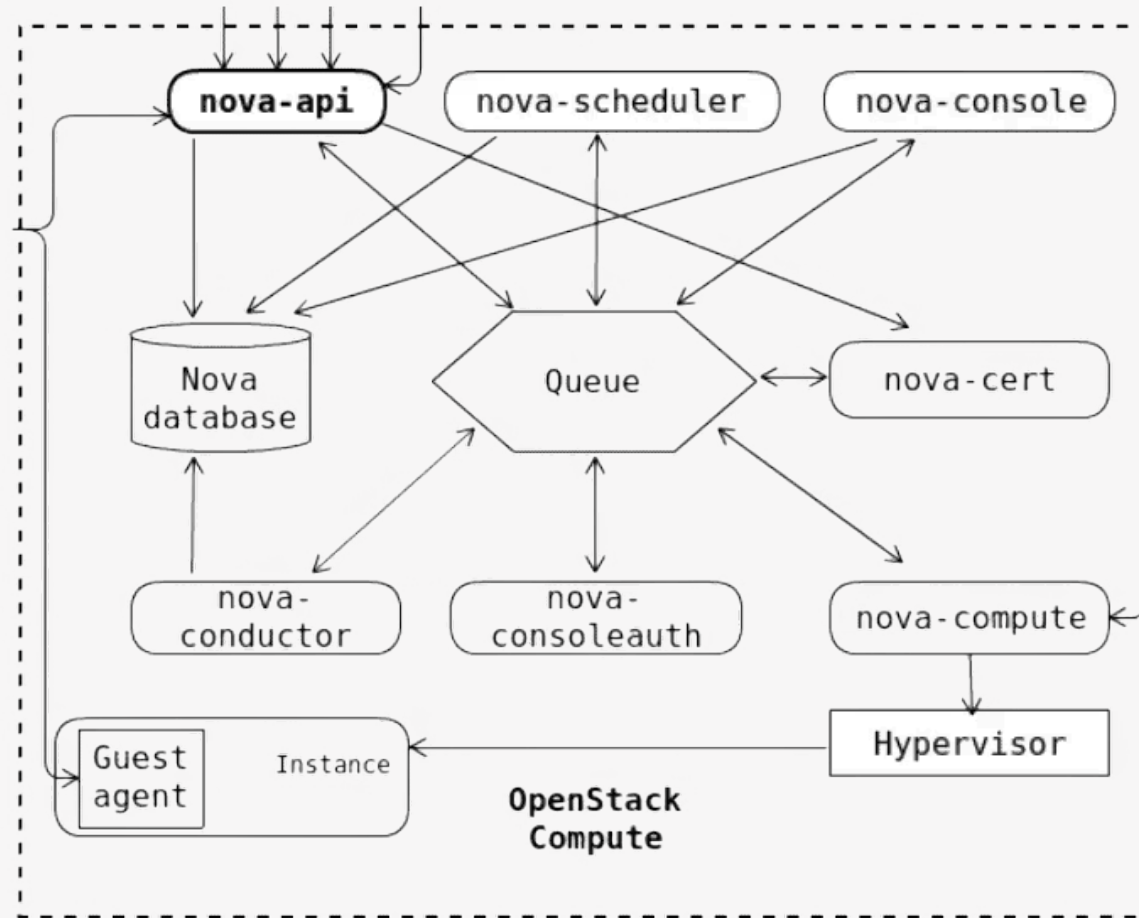
IaC and Automation tools



Database & Messaging

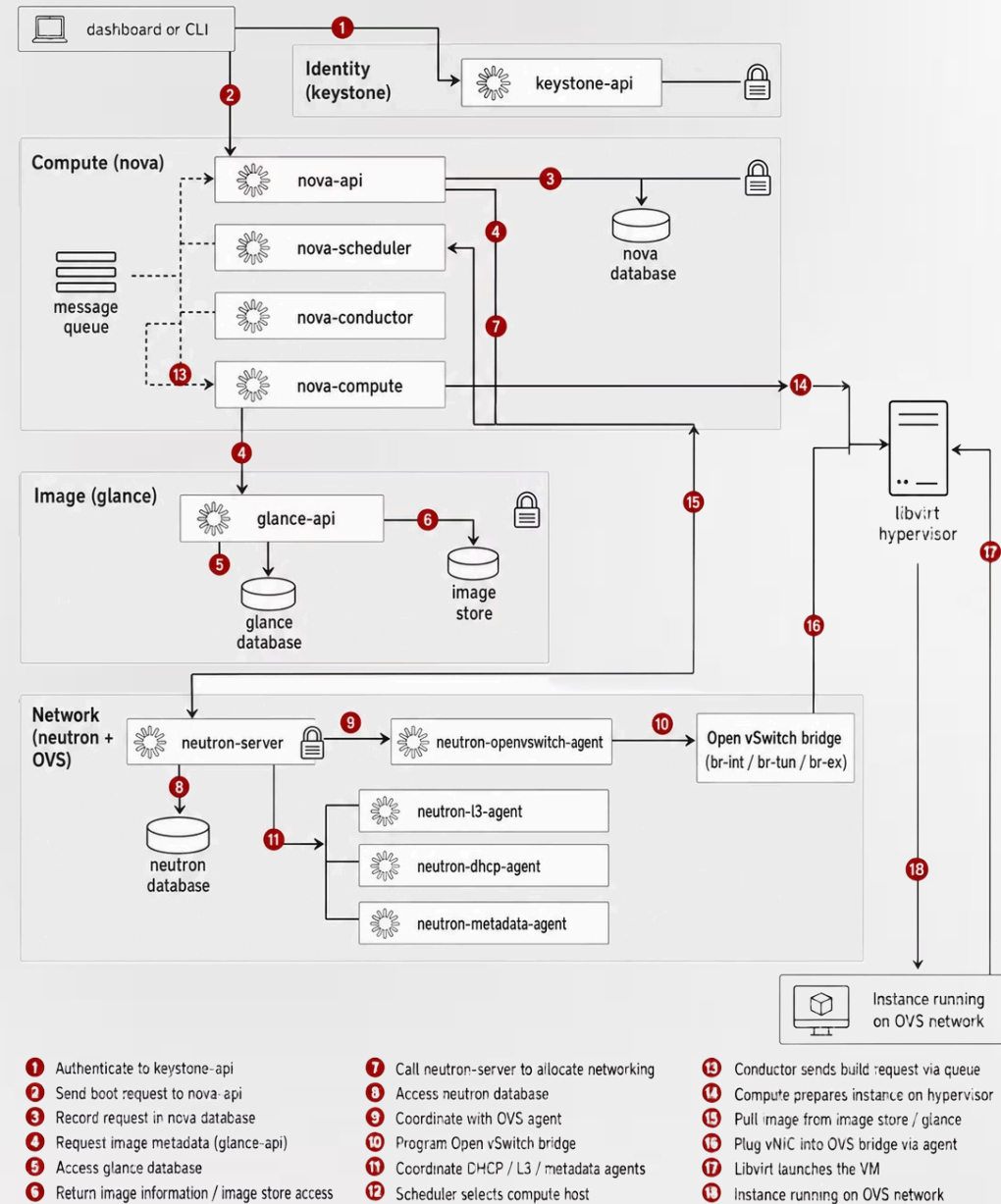


Nova - Compute Service



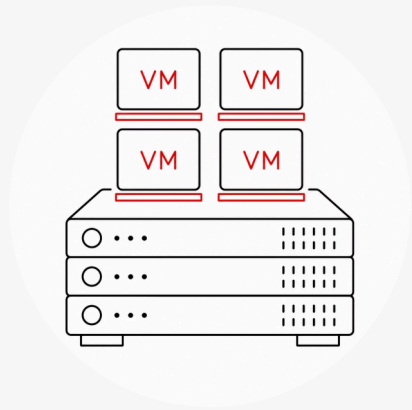
VM Creation Workflow

Create OpenStack Instance (OVS Network)

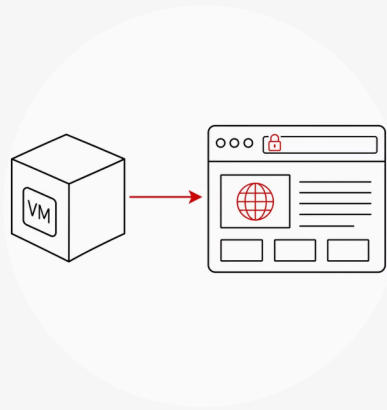


Virtualization

Key concepts and capabilities of virtualization in cloud computing



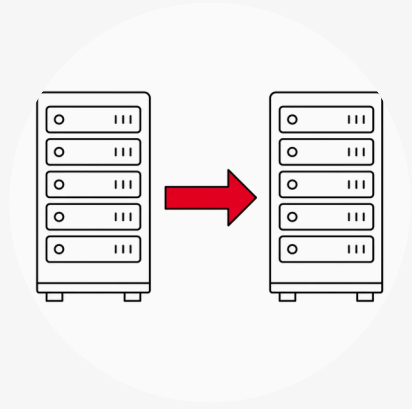
Density



Application Isolation



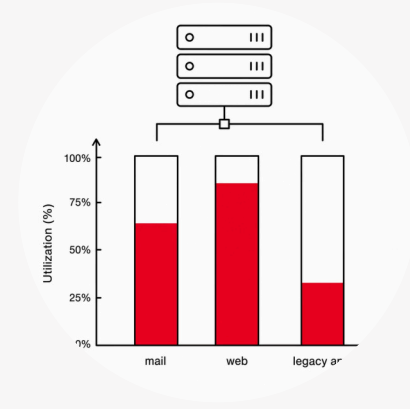
Snapshots



Failover & HA

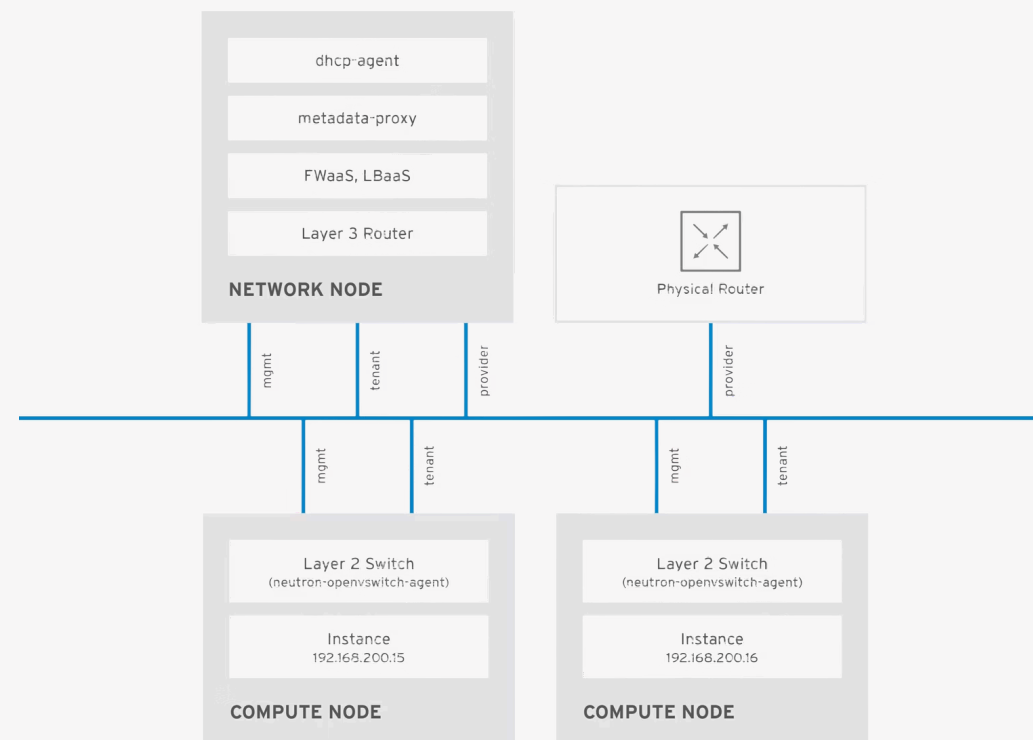
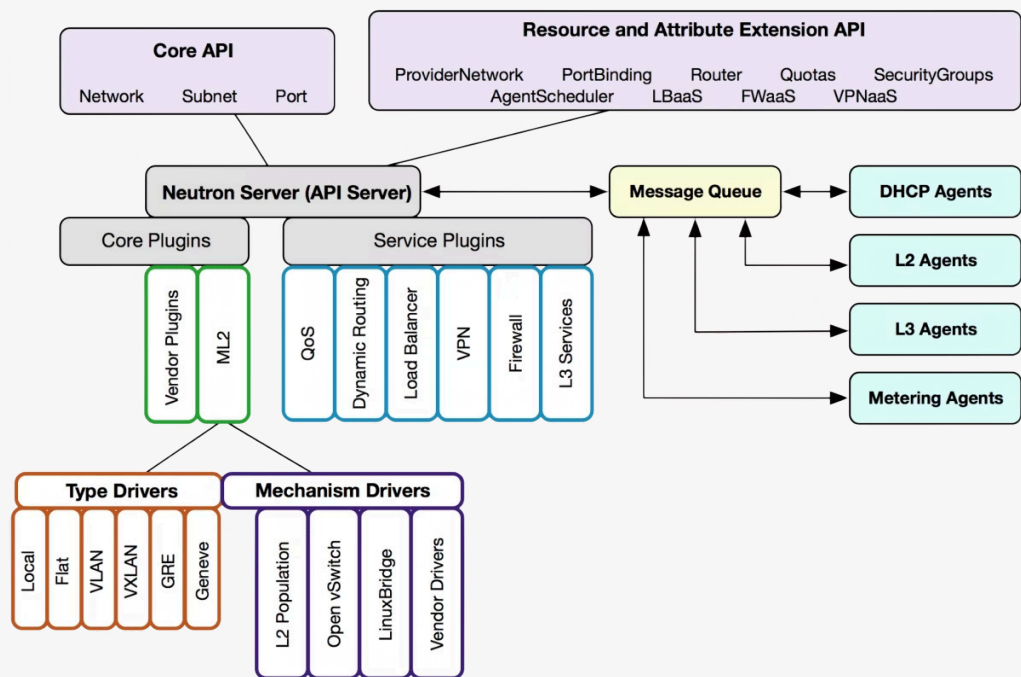


Migration

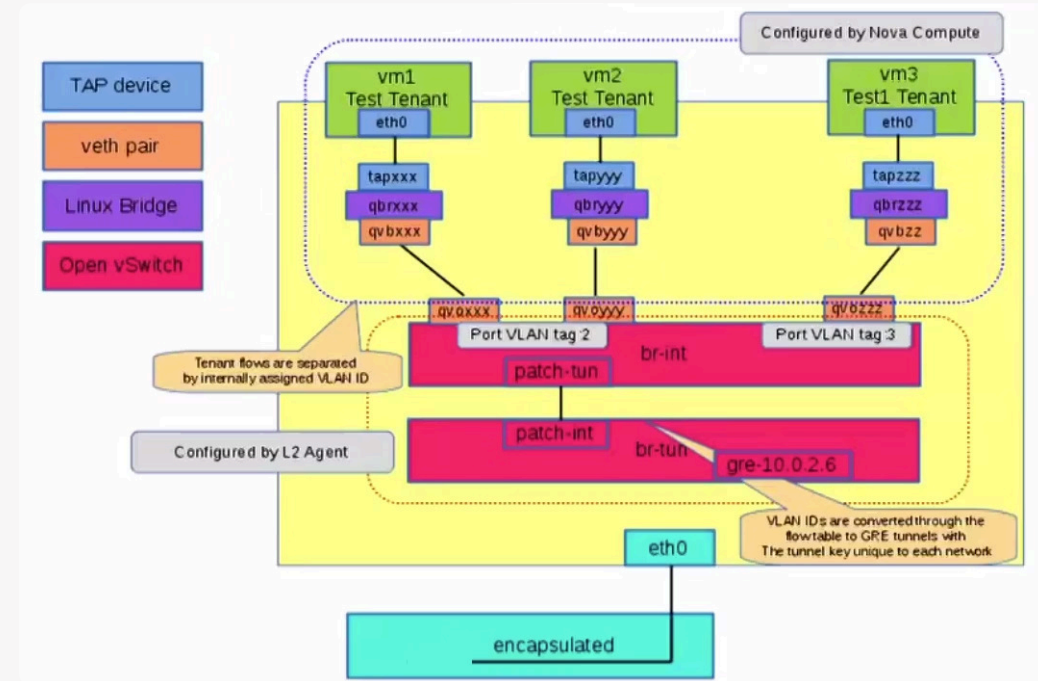
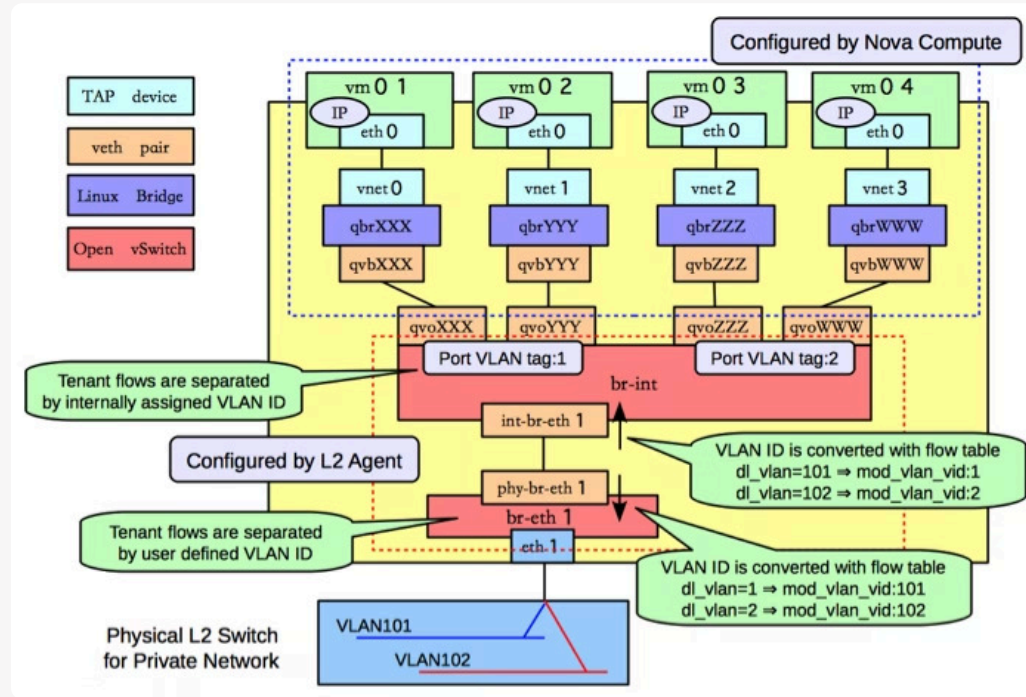


Efficient Utilization

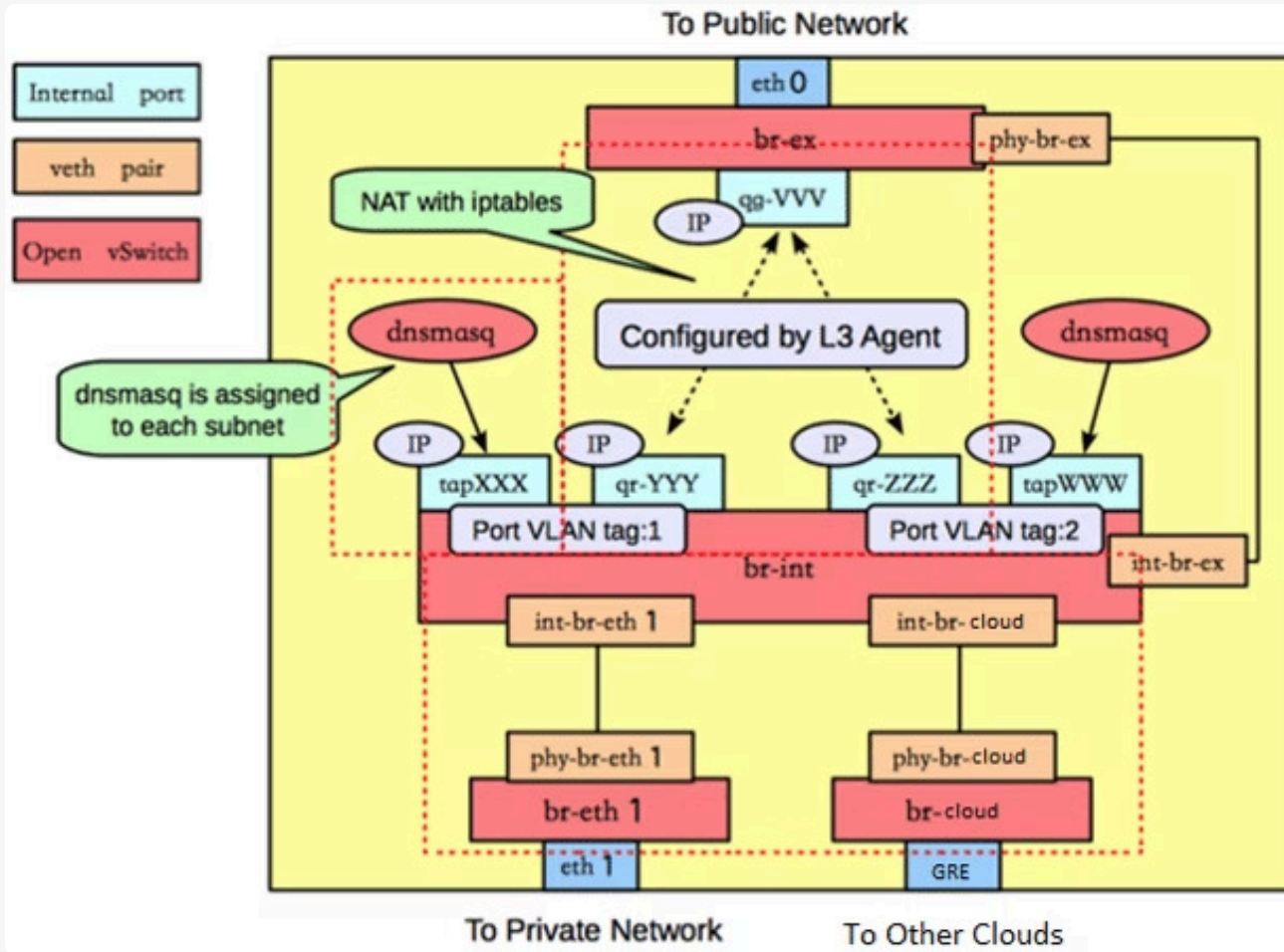
Neutron - Networking Service



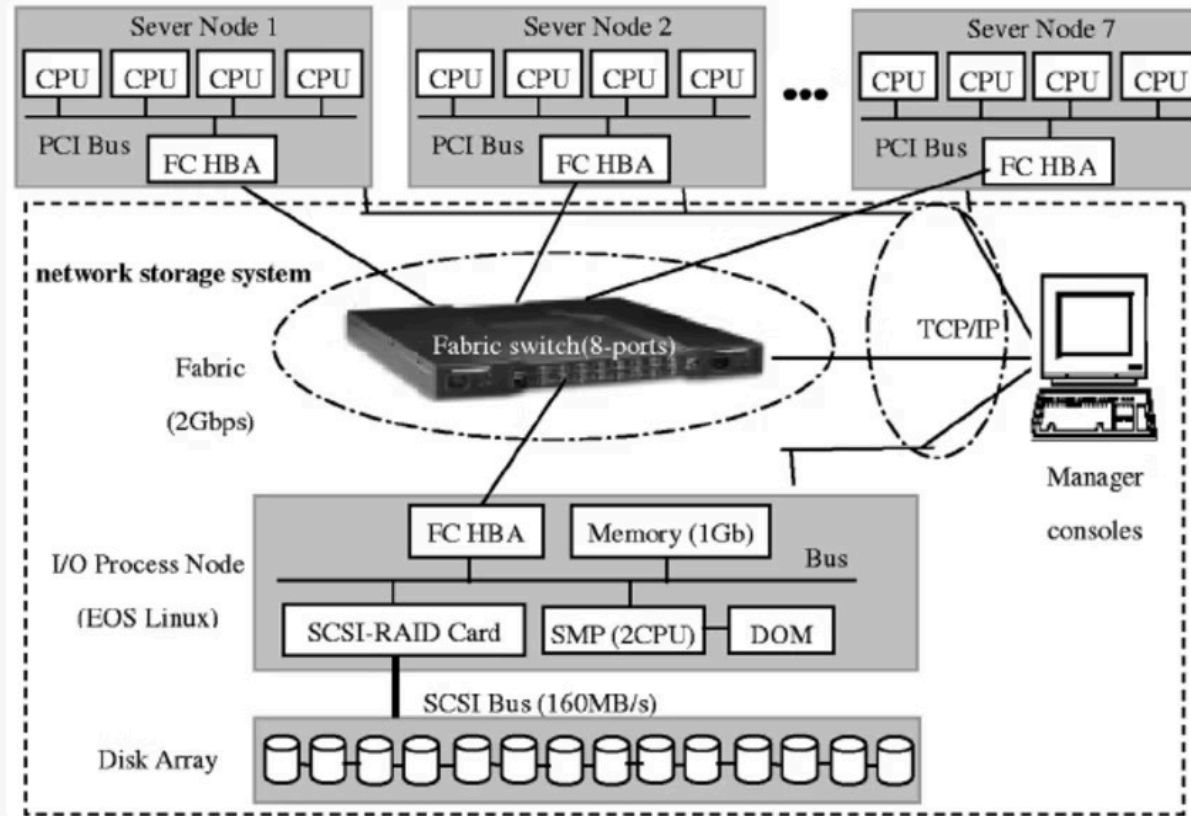
Neutron - Isolated Networking For Tenant VMs



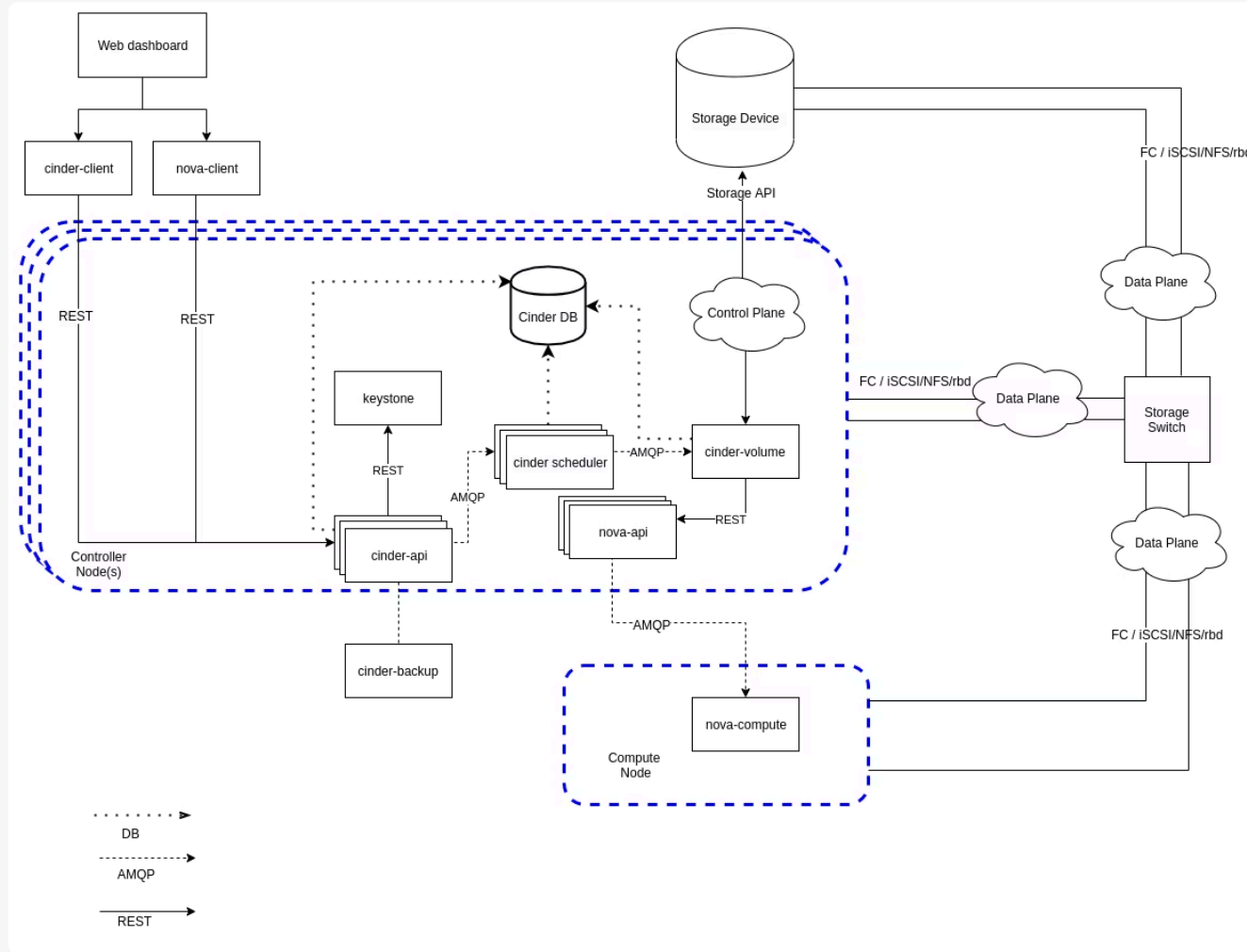
Neutron - DHCP And Routing



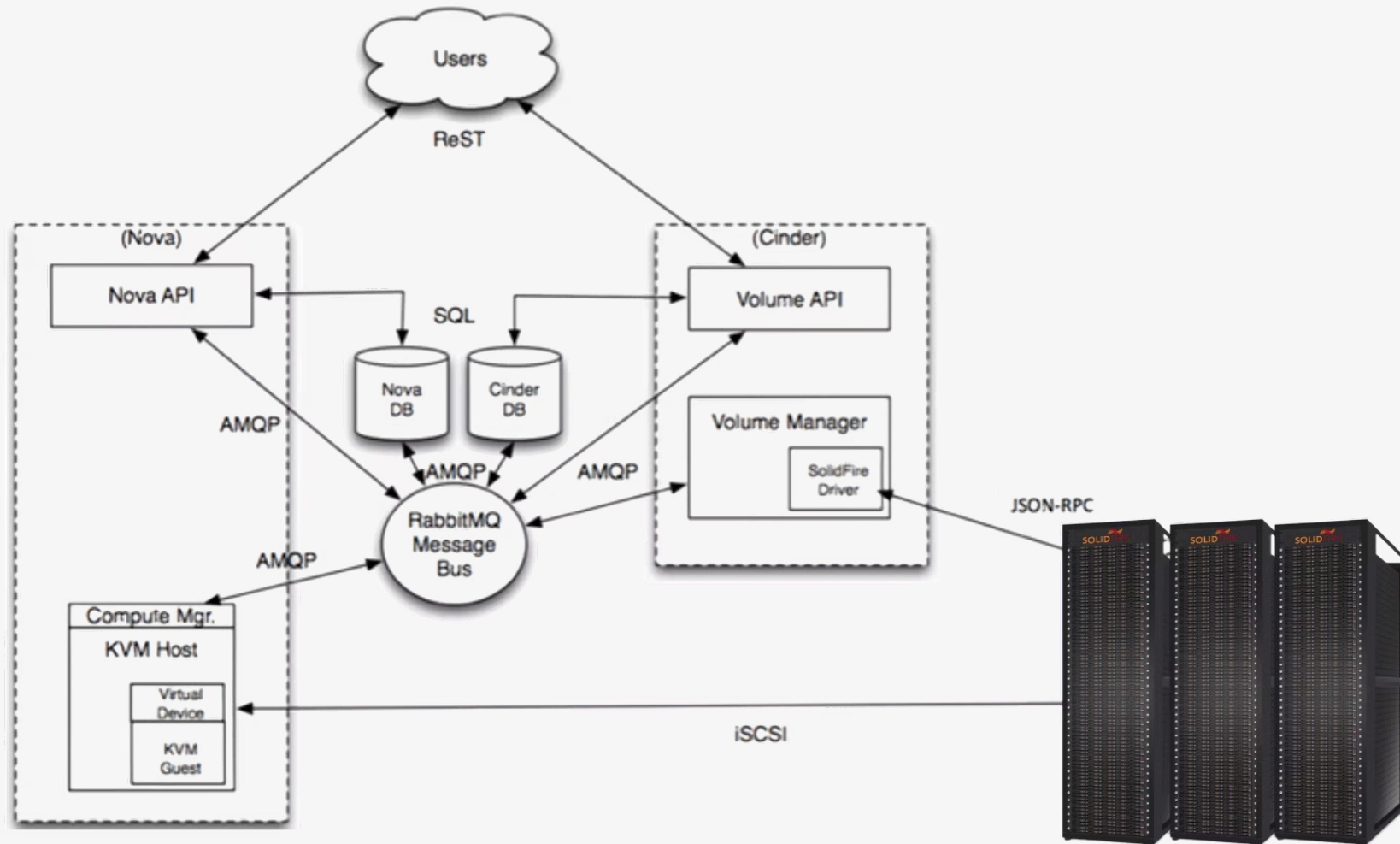
Block Storage Service - Cinder



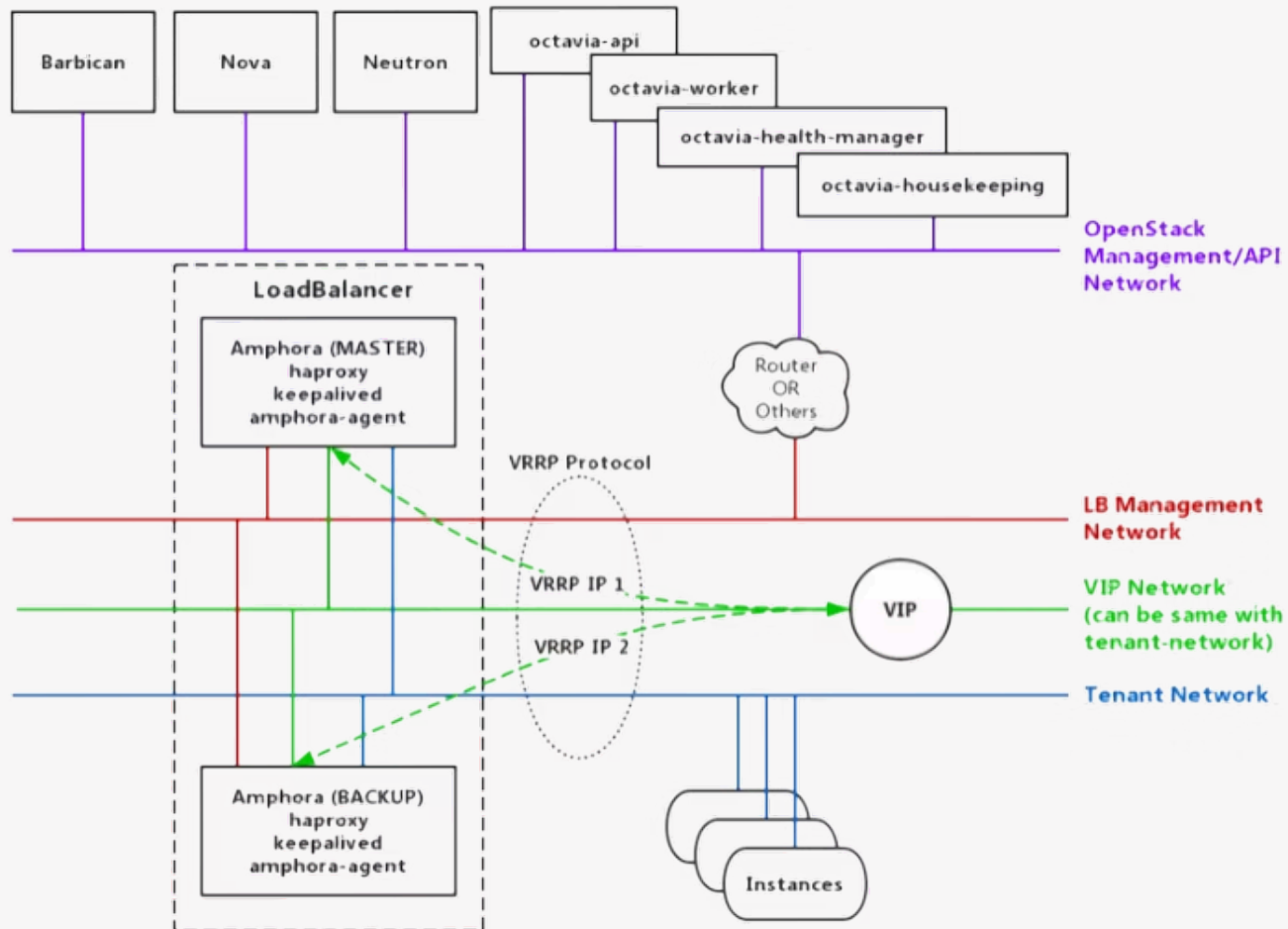
Block Storage Service - Cinder



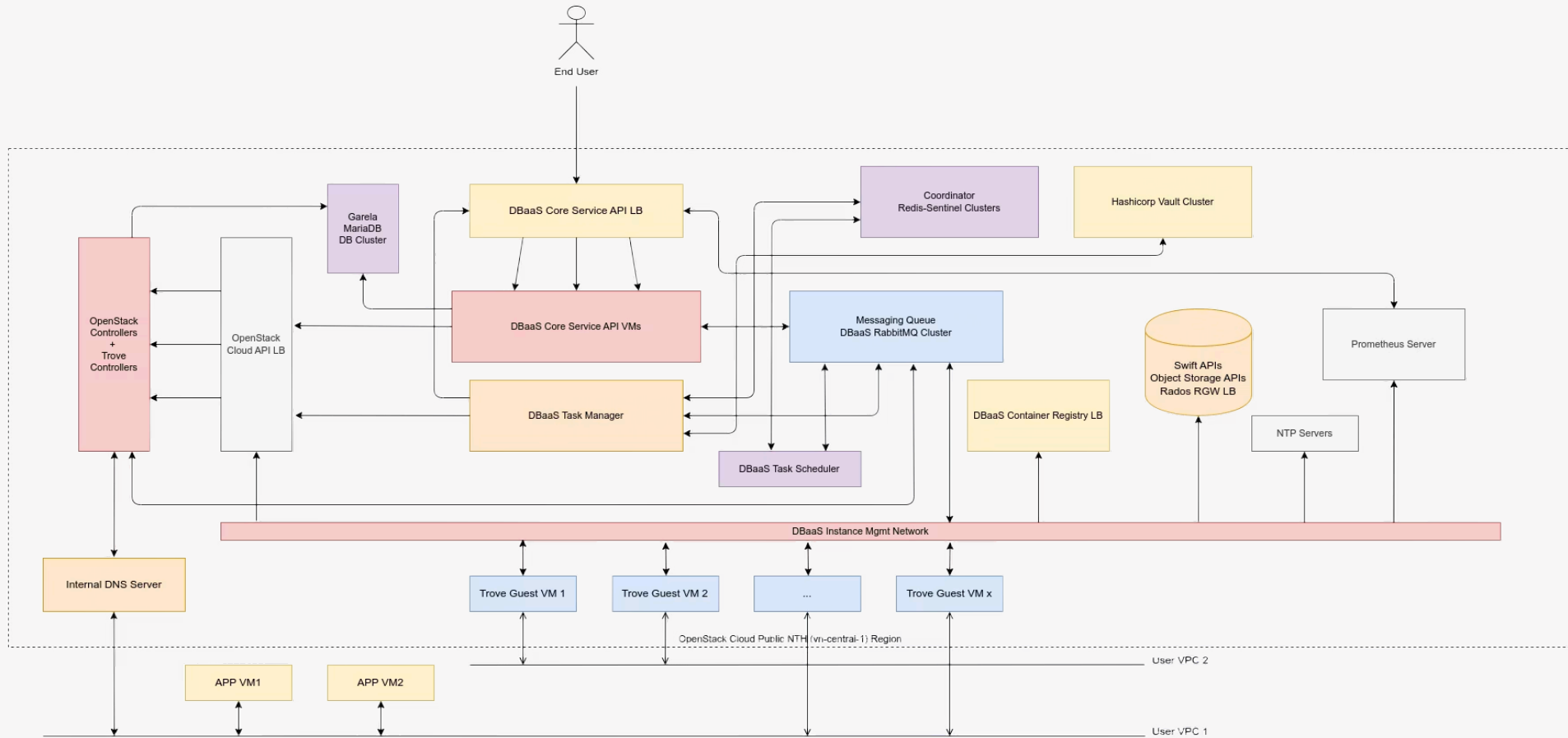
Cinder - Block Storage Volume Provisioning



Support Services: Octavia



Support Services: Database As A Service



Other Important Services

- Manila - File Sharing Service
- Barbican - Key Manager Service
- Designate - DNS Service
- ...

Next Steps?

**If this is your passion
build a cluster for yourself.**

viettel Cloud